MASTER FUNCTIONAL SKILLS

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AQA · LEVEL 2 · MATHEMATICS

AQA Functional Skills Mathematics Level 2

January 2023 - Paper 2 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-FS-Maths-L2-Paper-2-January-2023

100% Built from Empty Question Paper + Official Mark Scheme

Verified by Tutor Abdul · MasterFunctionalSkills.co.uk

HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	Master Functional Skills (Model Exemplar)
Centre Name:	Master Functional Skills (MFS01)
Lead Tutor:	Abdul · MasterFunctionalSkills.co.uk
Edition:	2026 Clean Empty-QP Rebuild (Zero Collisions Verified)

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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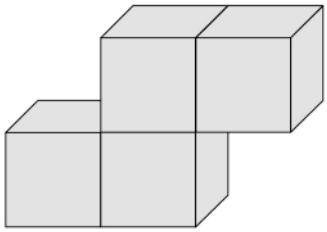
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Section A

Answer **all** questions in the spaces provided.

1 Here is a solid made from four cubes.



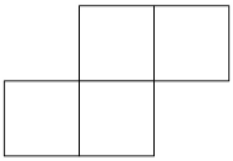
Q1 Worked Solution:

Two million three hundred (and) seven thousand (and) forty nine

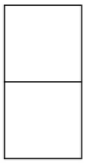
[B1] [B1]

Circle the plan view of the solid.

[1 mark]







2 Write the number 2 307 049 in words.

[1 mark]

Q2 Worked Solution:

Alternative method 1

0.614 with 0.121 and 0.114 seen

0.614 with 0.121 and 0.621seen

0.614 with 0.114 and 0.386 seen

[B1] [B2]

Do not write outside the box

3 Which is closer in value to 0.5

0.379 or 0.614 ?

You **must** show your working.

[2 marks]

Q3 Worked Solution:

0.379 + 0.614

or 0.4965

0.614 with 0.4965 seen

[M1] [A1]

Answer _____

4

Surface area of a sphere = $4\pi r^2$

Work out the surface area of a sphere with a radius of 2.3 cm

State the units of your answer.

[3 marks]

Q4 Worked Solution:

cm2

[B1]

Answer _____

Do not write outside the box

6 The table contains grouped data.

x	**Frequency**	**Mid-point**	
$0 < x \leqslant 20$	7	10	
$20 < x \leqslant 40$	9	30	
$40 < x \leqslant 50$	4	45	
	Total = 20		

Q6 Worked Solution:

Alternative method 1

÷ + Work out an estimate of the mean of x.

[3 marks]

29 (7 1) or 3(...)
29 – their 3 or 26
their 26 4.99 or 129.74
x +
29 4.25 3.90 or 127.15
(£)129.74 and (£)127.15 and Bags
and stuff
[M1] [M1] [M1] [A1]

Answer _____

12

Turn over for Section B

Do not write outside the box

Section B

Answer **all** questions in the spaces provided.

7 **School office**

Ashley works in a school office.

7 (a) Ashley needs 29 book bags.

He sees these offers online.

Q7(a) Worked Solution:

(£)129.74 and (£)127.15 and Bags and stuff

Best book bags	**Bags and stuff**
Book bags £4.99 each	Book bags reduced to £4.25 each
For every 7 you buy you get an extra 1 free	Delivery charge £3.90 per order
Free delivery	

Alternative method 1

π 3 Ashley wants the cheapest total price for the 29 bags and delivery.

Which offer should he choose?

You **must** show your working.

[5 marks]

× 72 × ×

25

[2307, 2309.4]

× (÷

40 their [2307, 2309.4] 1000)

or [92 280, 92 376] or [92. 2, 92 .4]

× (×

13 6 1000) or 78 or 78 000

[A1] [M1] [M2] [A1] [M1] [M1]

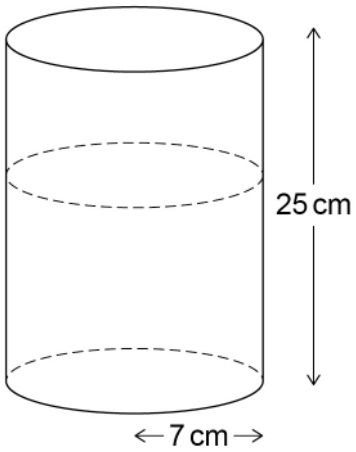
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7 (b) Ashley needs to put drinking water into 40 jugs before break time.

Each jug

is a cylinder with radius 7 cm and height 25 cm

will be $\frac{3}{5}$ full.



Water flows out of the tap at 13 litres per minute.

$1000\,cm^3 = 1$ litre

Can Ashley put the right amount of water into all 40 jugs in 6 minutes?
You **must** show your working.

[6 marks]

11

Do not write outside the box

8 Lifeguard

Talia is a lifeguard at a swimming pool.

8 (a) One morning, Talia checked the water temperature in the pool each hour.

Time	8 am	9 am	10 am	11 am
Temperature in °F	75	80	81	87

Q8(a) Worked Solution:

9 (am) and 26.6...

The formula to convert Fahrenheit (F) into Celsius (C) is

$$C = \frac{5}{9}(F - 32)$$

9 (am) and 26.7

[A1] What time was the first check where the water temperature was above 26°C?

You **must** show your working.

[3 marks]

Answer _____

Do not write outside the box

8 (b) There are 90 people in the pool.

number of adults : number of children = 3 : 2

One lifeguard is needed for every 10 children.

How many lifeguards are needed for the children in the pool?

[4 marks]

Answer _____

7

Turn over for the next question

Do not write outside the box

9 Hedgehogs

Aahad works for a charity that rescues hedgehogs.

9 (a) The charity has four main costs each month.

	Cost per month
Straw	£25
Feed	£39
Electricity	£68
Vets	£235

From next month the costs are changing.

- Straw will be free from a local farm
- Feed is increasing by $\frac{1}{6}$
- Electricity is increasing by £324 a year
- The cost of the vets will be £3090 a year.

The charity receives a new monthly donation of £30

Is the new donation enough to pay for the **increase** in total monthly costs?
You **must** show your working.

[5 marks]

Question 9(a) · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

Q9(a) Worked Solution:

45.5(0)
68 + 324 12 or 95
3090 12 or 257.5(0)
their 45.5(0) + their 95 + their
257.5(0) or 398
257.5(0) – 25 – 39 – 68 – 235 or 31
398 and 367 and No
31 and No
or 45.5(0) 12) or 546
× + +
68 12 324 or 816 324 or
1140
[M1] [M1] [A1] [M1] [A1] [M1]

Do not write outside the box

9 (b) The charity needs a new building for the hedgehogs.

The new building will cost £250 000

The charity invested £245 000 in a bank account 2 years ago.

The account pays 1.2% compound interest each year.

Is there enough money in the bank account to pay for the building?
You **must** show your working.

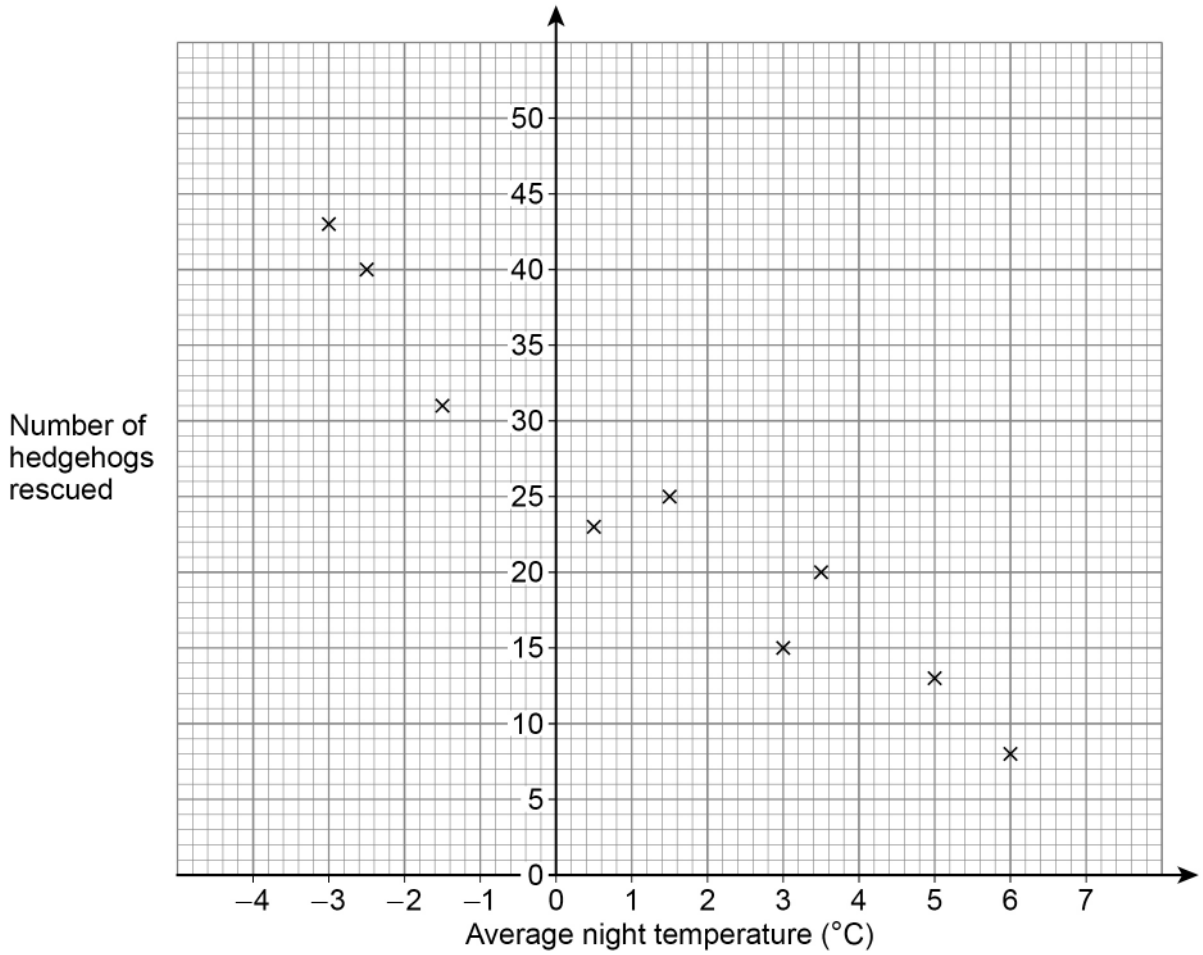
[3 marks]

Question 9 continues on the next page

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Each week Aahad records the number of hedgehogs rescued and the average night temperature that week.

The scatter diagram shows information for the last 9 weeks.



9 (c) Describe the relationship between average night temperature and number of hedgehogs rescued.

[1 mark]

Do not write outside the box

9 (d) In week A the average night temperature was $-2^{\circ}C$

In week B the average night temperature was $4^{\circ}C$

Use the scatter diagram to estimate how many **more** hedgehogs were rescued in week A than in week B.

You **must** show your working, some of which should be on the diagram.

[4 marks]

Answer _____

13

Turn over for the next question

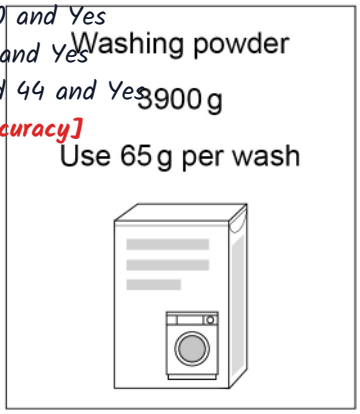
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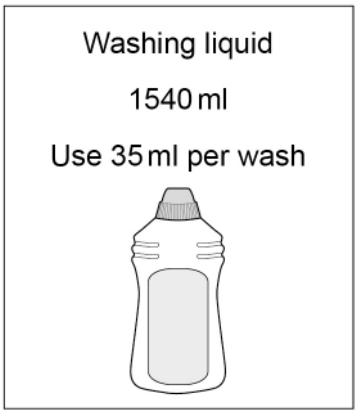
10 Housework

David is doing housework.

10 (a) David is washing clothes.

Q10(a) Worked Solution:
58.(...) and 60 and Yes
14.(..) and 16 and Yes
[45.9, 46] and 44 and Yes
[Method & Accuracy]





David says,

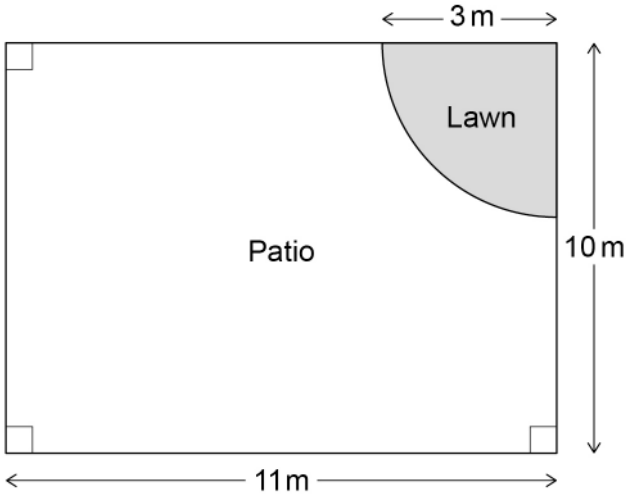
"I can do **at least** 32% more washes with the powder than the liquid."

Is he correct?
You **must** show your working.

[5 marks]

Do not write outside the box

10 (b) David is going to clean the patio.



Not drawn accurately

The lawn is a quarter circle with radius 3 m

Each bottle of cleaner

cleans 15 m^2 of patio

costs £1.95

Work out the cost of the bottles of cleaner that David needs to clean the patio.
You **must** show your working.

[6 marks]

Answer £ _____

Question 10 continues on the next page

Do not write outside the box

10 (c) David makes a cleaning spray using vinegar and water.

The cleaning spray needs to be

700 ml in total

45% vinegar.

David has put **half a pint** of vinegar in the spray bottle.

How much **more** vinegar and how much water does he need to add?
Use 1 pint = 568 ml
State the units of your answers.

[6 marks]

Vinegar _____

Water _____

17

END OF QUESTIONS

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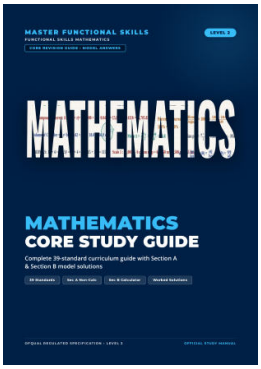
Question number	**Additional page, if required.** **Write the question numbers in the left-hand margin.**
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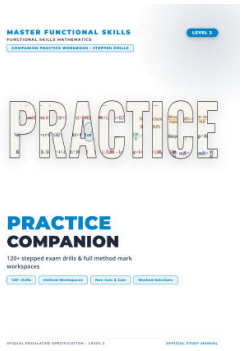


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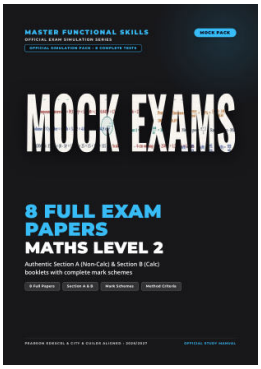


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